



UCSC

University of Colombo, Sri Lanka

University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester II— UCSC AY20 [held in March/April 2024]

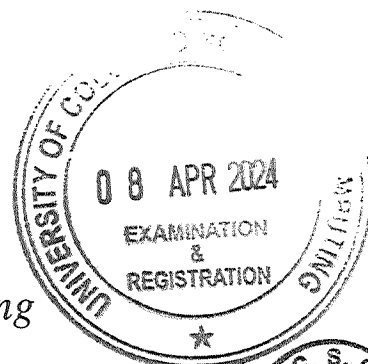
SCS1214 — Operating Systems I

(2 Hours)

Answer All Questions

Number of Pages = 12

Number of Questions = 4



To be completed by the candidate

Index Number

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Important Instructions to candidates:

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number **CLEARLY** on each and every page of this Question paper.
- This paper consists of 4 questions in 12 pages (including the Cover Page).
- Answer **ALL** questions.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are not allowed.
- Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate

To be completed by the examiners

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1. (a). The following programs are compiled and executed on a machine running Linux. Assume that all the required header files are included. Also, assume that all the `fork()` system calls returned successfully.

- i. What is the output of the program given bellow?

```
int main()
{
    int x;
    x=!fork();
    x=!fork();
    printf("%d", x);
}
```

[5 marks]

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- ii. What is the output of the program given bellow? Justify your answer.

```
int main()
{
    if (fork()==fork())
        printf("X\n");
    else
        printf("Y\n");
    return 0;
}
```

[6 marks]

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iii. What is the output of the program given bellow? Justify your answer.

```
int main()
{
    int x=2;
    while(fork())
        x--;
    printf("%d", x);
}
```

[6 marks]

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(b). A system has r instances of the resource type R and three processes, P_0, P_1, P_2 , that require R . The maximum requirements of R for each process and the current allocations at time t_0 are given in the following table. If a process is allocated its maximum requirement of a resource, it is guaranteed to finish and release its allocated resources within a finite amount of time.

	Maximum Need	Current Allocation
P_0	n_0	c_0
P_1	n_1	c_1
P_2	n_2	c_2

It is given that,

$$n_0 = r - c_2$$

$$n_1 = r - c_0 - c_2$$

$$n_2 < r$$

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- i. Define the term *safe state* with reference to deadlocks?

[3 marks]

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- ii. Is the above system in a *safe state* at t_0 ? Justify your answer.

[5 marks]

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2. (a). Which CPU ring levels in *x86* systems are used by the Linux operating system? Describe the purpose of each ring level in Linux.

[4 marks]

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- (b). i. The following program running on a Linux system creates two processes.

```
int x;
int main()
{
    x=2;
    if (fork())
        printf("%d\n", ++x);
    else
        printf("%d\n", --x);
}
```

Is there a **critical section** in this program? If there is a critical section, suggest a method to protect it. Justify your answer.

[5 marks]

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- ii. A program P has the statement X. Multiple processes execute the program P concurrently. At most 10 processes are allowed to execute the statement X concurrently. Write

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a code segment of P, in C-like pseudocode, to use semaphores to ensure that at most 10 processes are allowed to execute the statement X concurrently.

[5 marks]

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- (c). An operating system on a 32 bit system uses a single level page table. The memory is byte addressable. There are p entries in a page table. A page x that contains an address a of a process is mapped to the frame f .

- i. What is the size of a page in this system in bytes?

[3 marks]

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- ii. What is the physical address that maps to the address a ?

[3 marks]

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- iii. Assume that the page table only contains page-to-frame mappings and no other information. What is the minimum amount of memory required to store the page table? Justify your answer.

[5 marks]

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3. (a). Consider the following questions based on First-Come First-Served (FCFS) scheduling.

i. List two (2) main advantages and two (2) drawbacks of FCFS scheduling.

[4 marks]

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ii. How does FCFS scheduling affect turnaround time?

[2 marks]

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iii. Can FCFS scheduling lead to the “starvation” of certain processes?

[2 marks]

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iv. Which condition does FCFS scheduling consider for scheduling? For example, can you consider process priority or burst time?

[2 marks]

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- v. What is the name of the phenomenon that can result in increased turnaround times for short processes (in terms of burst time) and can impact the overall throughput of the system ?

[2 marks]

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- (b). Consider a priority scheduling algorithm running with three (3) processes - P1, P2, and P3 with the respective priority values as follows:

- Process P1 has a priority of 6.
- Process P2 has a priority of 8.
- Process P3 has a priority of 4.

Assuming higher numerical values indicate higher priority, answer the following questions:

- i. What is the initial order of processes in the ready queue based on their priorities?

[2 marks]

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- ii. If the processes arrive in the order P2, P1, P3, what will be the order of execution of processes after the arrival of each process?

[2 marks]

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- iii. During the current execution, the priority of P3 is increased to 7. How does this affect the order of execution (in terms of execution order) ?

[2 marks]

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- iv. If P2 is currently executing, and P1 with a higher priority of 9 arrives, what would be the new execution order of the processes?

[2 marks]

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- v. If each process takes a burst time of 12 milliseconds, what is the total **turnaround time** and **waiting time** for each process when considering part iii execution order?

[5 marks]

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4. (a). What is a “warm cache” ? How does **cold start penalty** impact the performance of the system? List a precaution taken by hardware vendors to reduce such performance penalties during context switches.

[6 marks]

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- (b). List two (2) advantages of using **single shared queue** over **private queues** in multi-programming.

[2 marks]

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- (c). List two (2) disadvantages of **linked allocation scheme**?

[2 marks]

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- (d). Why does system developers had to introduce variations of pages tables eg: hierarchical, hashed or inverted page tables ?

[2 marks]

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- (e). List two (2) algorithms that can be used to allocate page frames to a process by an operating system ?

[2 marks]

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- (f). What is the difference between shared and exclusive locks in file system ?

[2 marks]

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- (g). List two (2) features of tree-structured directory scheme in file system structure.

[2 marks]

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- (h). List two (2) techniques used by the file system to manage free space in the system.

[2 marks]

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(i). What are whole system VMs ? Provide an example for whole system VM.

[3 marks]

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(j). How does a hypervisor retain control of resources when serving multiple VMs ?

[2 marks]

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